Embedded Systems Project Showcase

Integrated Computer Engineering (Batch 23)

Scheduling and Valve Control to Cut Steam Peaks and Idle Waste in Solid Tyre Curing Presses

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1 Application scenario

- Sri Lanka is the world's largest solid tyre maker: nearly 25% of global demand.
- Presses have only a manual on/off steam valve: no sensor, no controller.
- Replacing presses does not meet a real ROI, so this project suggests a retrofit structure to overcome both problems.



Fig. 1: A curing press at a visited plant

Problem 1: Demand spikes

All presses are switched on together at shift start. Steam demand spikes and the boiler burns extra fuel.

Problem 2: Idle steam waste

At breaks, shift changes and overnight the valves stay open. Idle presses keep drawing steam.

2 Literature review

Work	What it does	Gap this work fills
Pentakota 2023	Energy analysis of a bicycle tyre press	Shows where energy goes, no scheduling or control
Tshimbiluni 2016	Mould warm up in tyre plants	Not combined with peak steam scheduling
Cancela 2018	MILP for tyre curing scheduling	Minimises makespan, not peak steam demand
Sunil Kumar 2022	Peak minimising MILP (buildings)	Not for curing presses or steam
Baldea 2014	Scheduling with process control (review)	General process industries, not curing presses
This work	**Peak steam MILP + valve control**	**Retrofit for manual presses, tested on twin and bench**

3 System design architecture

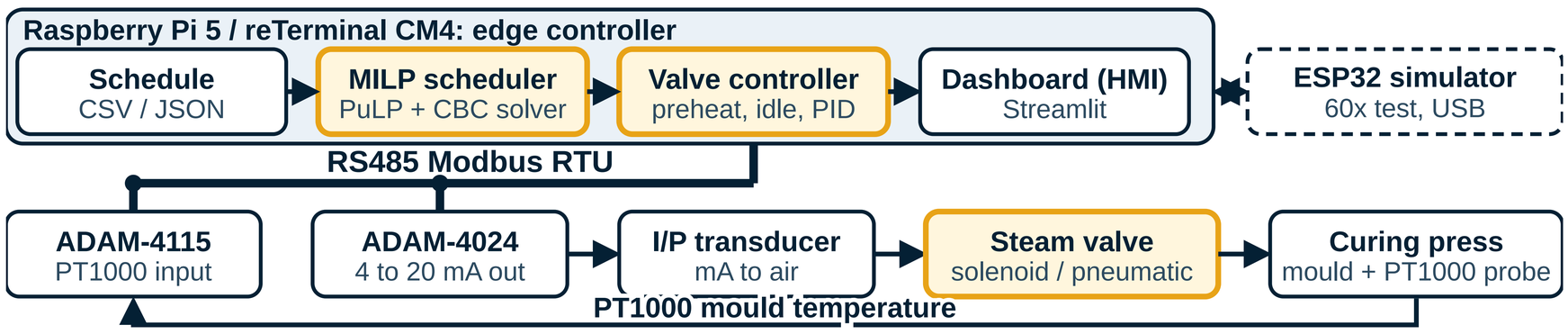


Fig. 2: The controller sends valve commands over Modbus; the PT1000 probe feeds back mould temperature.

4 Bench prototype

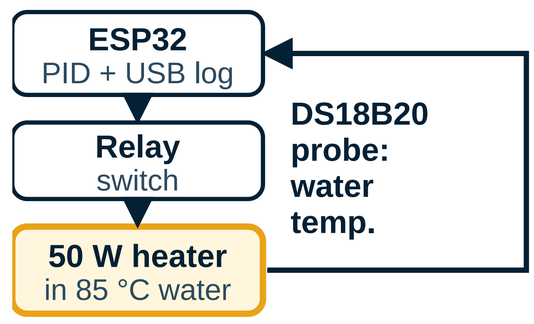


Fig. 3: Bench model. A 50 W heater in 85 °C water stands in for steam.

5 MILP scheduler

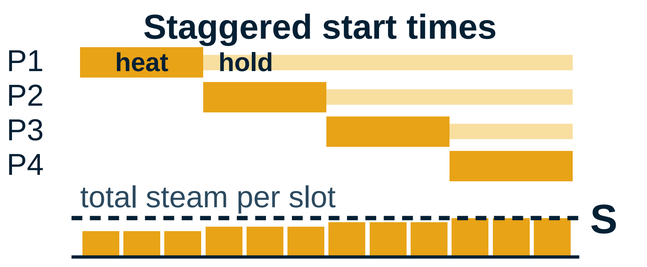


Fig. 4: MILP start times for 4 presses. Heat ups are spread out, so the peak S stays low.

- Time split into 5 min slots
- One start slot per press, inside its start window
- Solved with PuLP + CBC, 15 s limit

minimise S

S = the peak steam draw

$$\sum_t x_{p,t} = 1$$

each press starts once

$$\sum_p s_p(t - \text{start}_p) \le S$$

total steam in every slot stays under S

6 Results

58% to 63% lower peak steam (twin)

20.4% less idle gap steam (twin)

0.04 °C preheat error (twin)

52% less idle gap energy (bench)

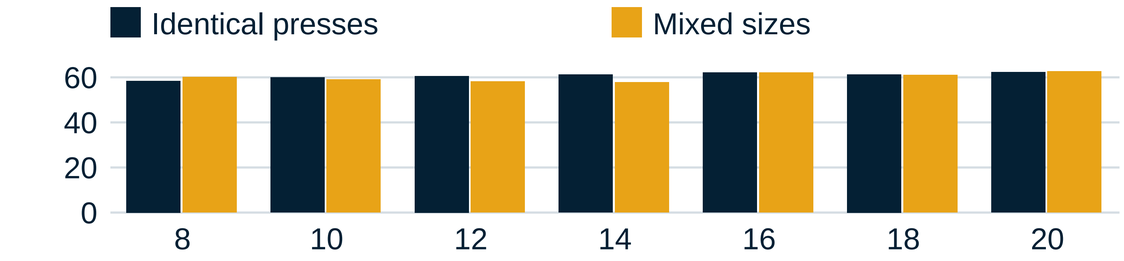


Fig. 5a: Peak steam cut (%) vs all presses starting together (twin).

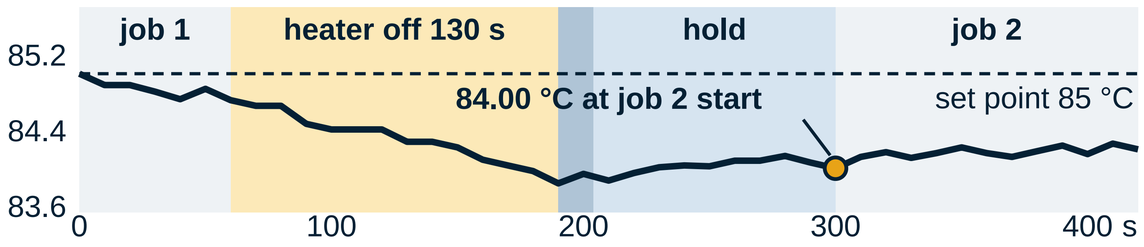


Fig. 5b: Bench idle gap: heater off 130 s, 84.00 °C at job 2 start.
Single run. Baseline energy estimated. Smart run started 0.8 °C warmer.

7 Conclusion

- **Digital simulation (twin):** peak steam draw 57.85% to 62.65% lower for 8 to 20 presses; idle gap steam 20.4% lower (1.62 kg to 1.29 kg); preheat error 0.04 °C vs 11.91 °C.
- **Bench:** PID overshoot 0.5 to 0.7 °C (on/off: 1.5 °C); preheat within 0.75 °C (fixed lead: 27.00 °C short); idle gap energy 52% lower.

8 Future work

- **Pilot on real curing presses**
- Full shift tests with real steam
- Tune anti windup and idle margin

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